Solutions to Chapter 8 Exercises:

Exercise 1 Solution

Simpler models like linear regression outperform deep models under stable or near-linear traffic patterns because they generalize better and avoid overfitting; LSTMs outperform when traffic is highly volatile or exhibits long-range temporal dependencies.

Exercise 2 Solution

In stable environments, statistical threshold detectors perform adequately and are more interpretable; deep autoencoders outperform when anomalies are subtle or nonlinear, but require more tuning and may overfit without enough diverse training data.

Exercise 3 Solution

Both scaling methods improve convergence; z-score normalization helps for models sensitive to variance, while min–max often produces faster convergence for neural networks. Without scaling, gradient-based training is unstable.

Exercise 4 Solution

Data augmentation (e.g., injecting noise, SMOTE-like synthetic samples) improves recall on rare attack classes and reduces class imbalance; however, excessive augmentation may distort patterns, causing overfitting or false positives.

Exercise 5 Solution

Quantized models reduce latency and power use significantly with minimal accuracy loss; FP32 yields slightly higher accuracy but is slower and consumes more energy, making quantization preferred for edge deployment.

Exercise 6 Solution

A learning rate of 0.1 often diverges, 0.001 converges well and balances speed and stability, and 0.00001 converges slowly and risks underfitting; improper learning rates cause oscillations or excessively slow progress.

Exercise 7 Solution

The regularized agent generalizes better across varied traffic patterns and topologies, avoiding overfitting to specific conditions; the unregularized agent performs well on training scenarios but poorly on unseen conditions.

Exercise 8 Solution

Small models underfit complex encrypted traffic patterns, large models overfit and require long training times; medium-sized architectures provide the best accuracy-efficiency trade-off with lower risk of overfitting.

Exercise 9 Solution

The RBF kernel usually outperforms the linear kernel due to nonlinear QoS relationships; optimal C values are typically mid-range (e.g., 1 or 10), balancing margin width and classification error.